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JGME-D-26-00298

"Generative Artificial Intelligence Use in Pediatric Residency Personal Statements"

Original Submission

Matthew A. Porter, B.S. (Reviewer 1)

Reviewer Recommendation Term:	Major Revision
Overall Reviewer Manuscript Rating:	60
Custom Review Question(s):	Response
Please rate the manuscript based on if the topic may be of interest to program directors or GME educators, researchers, or deans.	5=strongly agree
Please rate the manuscript based on if the conclusions are supported by data or evidence presented.	3=neutral
Please rate the manuscript based on if the manuscript needs additional statistical review.	3=agree
Comments to Editor:	
Is there a financial or other conflict of interest between your work and that of the authors? YES ___ NO ☒ If yes, please contact the Editor-in-Chief. The authors rely exclusively on Quillbot and ZeroGPT, consumer-facing tools with unvalidated accuracy in this context, deployed on applicant personal statements without applicant knowledge, a validated detection threshold, or an established appeals process. The authors' own 2022 baseline data confirm systematic false positives, and their demographic findings demonstrate disproportionate flagging of non-native English speakers. These findings raise ethical concerns that, in this reviewer's assessment, are insufficiently addressed by the current manuscript and warrant careful editorial consideration prior to any publication decision. This reviewer respectfully brings to the editor's attention a power dynamic concern that sits at the heart of this work: the manuscript, as written, risks positioning program directors as AI-empowered adjudicators of applicant authenticity, while the applicants themselves, many of whom are students with limited financial resources and unequal access to AI tools, have no knowledge that such screening is occurring, no recourse if flagged, and no representation in the design of the criteria being used to evaluate them. This asymmetry is compounded by the manuscript's complete absence of any engagement with purpose-built, GME-specific detection infrastructure, most notably the Thalamus GME Cortex platform, which was developed expressly for residency application screening with a published methodology and bias mitigation framework. A manuscript that equips program directors with consumer-grade detection tools while ignoring the institutional frameworks designed to protect applicants does not reflect the ethical standards the GME community should aspire to model. This reviewer would respectfully suggest that requiring major revision with a dedicated ethical framework, or soliciting an additional independent review from a medical education research methodology expert or GME equity scholar, represents an opportunity for JGME to signal clearly to its readership, and to the students who will be affected by these practices, that graduate medical education is committed to equity-centered implementation of emerging technologies. This is particularly salient for underrepresented and internationally trained applicants who may lack access to the same AI tools and institutional support as their peers, and who the authors' own data show are disproportionately flagged by the very instruments this manuscript implicitly endorses. Additional methodological concerns regarding the demographic anomaly in Table 1, manual data transcription validation, AI use threshold definition, and internal numerical discrepancies have been communicated directly to the authors in the open review comments. This reviewer remains open to reconsidering the manuscript following a major revision, but would strongly encourage the editor to weigh the equity dimensions of this work carefully in any interim publication decision.	
Comments to Author:	
General Comments to Authors	
This manuscript addresses a timely question regarding generative AI use in pediatric residency personal	

statements, and the authors are commended for assembling a large, multi-year dataset. However, significant methodological concerns preclude publication in its current form. Central to this review's concerns is the broader context in which this research sits: program directors are increasingly being positioned as both consumers and enforcers of AI detection, yet the tools available to them vary enormously in validated accuracy, institutional support, and ethical safeguards. The manuscript misses a critical opportunity to engage meaningfully with this landscape, and this gap is particularly difficult to reconcile given the public availability of purpose-built GME infrastructure at the time of this manuscript's preparation. Most notably, in November 2024 the AAMC and Thalamus formally announced the expansion of Cortex, an AI/ML-enabled holistic application review platform, to all ERAS-participating programs as a complimentary benefit beginning July 2025 for the 2026 ERAS season. This announcement predates this manuscript's submission. The AAMC has confirmed on its own institutional FAQ page (<https://www.aamc.org/services/eras-institutions/faqs-aamc-thalamus>) that Cortex uses artificial intelligence, machine learning, natural language processing, and optical character recognition to aggregate key application data in alignment with the AAMC's published principles for responsible AI in residency and fellowship selection. Cortex has been operational in some capacity since 2020 and has been used to screen over three million residency and fellowship applications at institutions throughout the United States. It was piloted in the 2025 ERAS season and was fully integrated and available to all programs by July 2025. A manuscript that aims to inform program director practice on AI use in personal statement review, submitted after the AAMC's formal announcement of this tool's universal availability, and without any acknowledgment of its existence, presents a gap in the literature review that is difficult to reconcile and should be addressed in revision. Equally important is the manuscript's failure to account for the AI tools most accessible to the very applicants under study. Microsoft Copilot is now natively embedded in Microsoft 365, including Microsoft Word, the most common tool used to draft personal statements, and is physically integrated into an increasing number of student-facing devices via dedicated Copilot hardware keys. This represents arguably the most widely available and frictionless AI writing assistant available to applicants today, yet it receives no acknowledgment in the Introduction, Methods, or Discussion. A study examining AI use in personal statements that does not account for the most ubiquitous AI writing environment available to students has a significant blind spot that must be addressed.

The specific comments below detail each concern with reference to the relevant pages and line numbers of the manuscript.

Specific Comments to Authors

Comment 1

Page 1, Lines 9-12 and Page 3, Lines 9-12

Although OpenAI's ChatGPT is arguably the most well-known AI tool across the medical education landscape, it is unspecific at best and misleading at worst to equivocate ChatGPT as a catch-all for AI. The abstract and introduction use "AI (i.e. ChatGPT)" as though these terms are interchangeable, which they are not. ChatGPT is one product from one company; AI is a broad field encompassing numerous tools, architectures, and use cases. This conflation risks biasing the reader's interpretation of the findings, misrepresenting the landscape of tools available to applicants, and limiting the generalizability of the study's conclusions. The authors should distinguish clearly between generative AI as a category and ChatGPT as a specific product throughout the manuscript.

Comment 2

Page 4, Line 14

OpenAI's GPT-4 model was released in March 2023, not 2022 as the authors have written. The factual release date for GPT-4 can be confirmed via OpenAI's own release announcement at <https://openai.com/index/gpt-4-research/>. The platform publicly released in November 2022 was ChatGPT, which at launch was powered by GPT-3.5, not GPT-4. This is not a minor distinction: the study's entire temporal framing depends on accurate characterization of which AI model generation was available during each application cycle. The authors should correct this error and clarify which model generation they believe applicants were using in each cycle year.

Comment 3

Page 6, Lines 19-21

The authors state that Quillbot and ZeroGPT were selected in part because they are "free, open-access platforms" with "good accuracy." While the intent to minimize cost barriers is understandable, this justification warrants further scrutiny on two grounds: the characterization of these platforms as free, and the adequacy of the accuracy data provided. Both Quillbot and ZeroGPT operate freemium models. While basic functionality is available at no cost, each platform offers paid tiers with expanded features and higher usage limits. The claim that these tools were selected because they are "free" may therefore be imprecise and should be clarified. More critically, the cost justification fails to account for purpose-built, validated GME infrastructure that is already available to all residency programs at no cost. The AAMC has confirmed that starting July 2025, all ERAS-participating programs receive complimentary access to the Thalamus Cortex platform (<https://www.thalamusgme.com/add-ons/cortex-application-screening>), which uses AI/ML, NLP, and optical character recognition to analyze application materials including personal statements, in alignment with the

AAMC's published principles for responsible AI in residency selection (<https://www.aamc.org/services/eras-institutions/faqs-aamc-thalamus>). Thalamus publishes methodology white papers for each AI feature through its AI Hub (<https://www.thalamusgme.com/ai-hub>), including a published methodology for the Academic Career Interest Badge, which uses a large language model to analyze personal statements for expressed and implied applicant characteristics (<https://www.thalamusgme.com/blogs/methodology-for-the-academic-career-interest-badge-in-cortex>). An independent peer-reviewed early evaluation of Cortex has been published in *Laryngoscope* (Erickson et al., 2026, DOI: 10.1002/lary.70386), providing a citable, independently assessed evidence base that the authors could have engaged with. While Cortex is not designed as a standalone AI-generation detector, it represents the standard of purpose-built, bias-conscious, and institutionally validated tooling available to the GME community, the existence of which the manuscript does not acknowledge at all. Additionally, the authors' institution (Yale School of Medicine) provides its community with free institutional access to several enterprise-grade AI platforms, including Anthropic's Claude, Google Gemini, and Microsoft Copilot, through Yale's AI Tools and Resources portal (<https://ai.yale.edu/yales-ai-tools-and-resources>). Yale's own AI research infrastructure, including the Aneja Lab (<https://github.com/Aneja-Lab-Yale>) and its contributions to open-source medical AI such as Meditron via the Yale-EPFL collaboration (<https://ai.meta.com/blog/llama-2-3-meditron-yale-medicine-epfl-open-source-llm/>), further underscores the breadth of validated AI tooling available to the authors' institution that was not considered in this study. The rationale for selecting consumer-grade freemium tools over these validated, institutionally supported, and purpose-built alternatives should be explicitly addressed in a revised Methods section.

Comment 4

Page 6, Lines 19-21

The claim that these platforms have "good accuracy" lacks the specificity required given the high-stakes nature of their application. Weber-Wulff et al., the very study cited as Reference 24 in this manuscript, evaluated 14 AI detection tools and found that "all scored below 80% accuracy and only 5 achieved over 70%," concluding that available detection tools are "neither accurate nor reliable" with a systematic bias toward classifying AI-generated output as human-written. Using the binary inclusive evaluation method, ZeroGPT achieved an accuracy of 74%, and only Turnitin reached the maximum of 81% across all tools tested. Critically, accuracy dropped sharply for hybrid writing scenarios: tools achieved only approximately 50% accuracy for AI-generated text that was manually edited, and approximately 26% accuracy for AI-generated text that had been machine-paraphrased, the very conditions most likely to appear in real residency personal statements. The paper explicitly warns that "reports provided by detection tools for AI-generated text cannot be used as the only basis for reporting students for cheating." In the context of residency application screening, where a false positive could unjustly stigmatize an applicant's candidacy, the phrase "good accuracy" is insufficient and unsupported even by the authors' own cited literature. The authors should directly acknowledge these limitations and the specific performance data of ZeroGPT and Quillbot as documented in Reference 24, and address whether a consensus-based approach aggregating multiple detectors was considered as part of the study design.

Comment 5

Page 6, Lines 19-23; Page 7, Lines 24-26; Figure 2, Page 21

The study reports AI detection as a continuous variable (mean AI%) throughout the analysis, yet Figure 2 and the associated results text present a binary outcome defined as "any amount of AI detection," a threshold of greater than 0%. This creates two distinct analytical problems. First, using greater than 0% as the threshold for "AI use" is methodologically unjustifiable given the known baseline false positive rates of both platforms; as the authors themselves acknowledge, 2022 statements written prior to the public release of ChatGPT were nonetheless flagged as containing AI-generated content, with a mean Quillbot score of 4.9% (SD 12.9) and ZeroGPT mean of 35.9% (SD 21.1) (Table 2, page 20). Any threshold that classifies these pre-ChatGPT statements as AI-generated is, by definition, capturing noise rather than signal. Second, the continuous mean AI% data presented in Table 2 treat a statement scored at 5% identically to one scored at 95% in group-level comparisons, despite these scores having fundamentally different clinical and operational interpretability. The authors should: (1) define and justify a meaningful minimum threshold for classifying a statement as AI-assisted (e.g., 20% or 50%), informed by the false positive rates documented in their own baseline 2022 data; (2) reanalyze and re-present Figure 2 using this validated threshold; and (3) consider stratified analyses across AI% bands (e.g., 1-20%, 21-50%, greater than 50%) to provide findings that are actionable for residency program directors.

Comment 6

Page 6, Lines 28-32; Page 7, Lines 19-22; Table 2, Page 20; Page 8, Lines 1-5

The authors state that applicants in the 2022 cycle are presumed not to have had access to ChatGPT, given that the platform was not publicly released until November 30, 2022, after the ERAS application deadline for that cycle had passed. This presumption is reasonable and effectively renders the 2022 cohort a natural negative control: any AI-generated content detected in 2022 personal statements cannot reflect actual applicant AI use and must therefore represent false positives. Table 2 reveals that both platforms produced substantial detection scores in this negative control year: Quillbot returned a mean of 4.9% (SD 12.9) and ZeroGPT returned a mean of 35.9% (SD 21.1) for 2022 applicants. The ZeroGPT figure is particularly concerning. A mean false positive rate of 35.9% on a cohort that could not have used AI represents a profound baseline noise level that fundamentally compromises the interpretability of all subsequent year-over-year

comparisons. The authors acknowledge this briefly in the Discussion, noting that "both detectors produced false positives," but do not adequately engage with its analytical implications. Specifically, the authors must address the following: (1) If ZeroGPT flags an average of 35.9% AI content in verified human-written 2022 statements, on what basis can any ZeroGPT score in 2023 or 2024 be interpreted as evidence of AI use rather than detector noise? (2) The year-over-year increase reported as a central finding of this study, from a ZeroGPT mean of 35.9% in 2022 to 51.5% in 2023 and 65.3% in 2024, must be interpreted against this baseline, yet no such adjustment or sensitivity analysis is performed. (3) Figure 2, which reports "any AI detection" as a binary outcome, is rendered uninterpretable if the baseline false positive rate already produces near-universal detection in human-written text. The authors should reframe their trend analysis using the 2022 false positive rate as a noise floor, conduct a sensitivity analysis excluding scores below the established baseline threshold, and revise their conclusions accordingly. Failure to do so means the manuscript's primary finding, that AI use is increasing, cannot be distinguished from the alternative explanation that detector sensitivity or bias increased over time.

Comment 7

Page 13, Lines 1-4 (Reference 11)

Angus et al.'s manuscript "Internal Medicine Residency Program Directors' Screening Practices and Perceptions About Recruitment Challenges" carries a potential citation date discrepancy. Depending on the citation convention used, this reference may reflect the online-ahead-of-print publication date of 2020 or the print issue date of 2021. The discrepancy was identified by visiting the DOI hyperlink provided by the authors (<https://doi.org/10.1097/ACM.0000000000003086>). The authors should verify and standardize their citation date convention consistently across all references in the manuscript. If following standard print issue citation convention, the reference should read: Acad Med. 2021 Apr;95(4):582-589. If following online-ahead-of-print convention, the reference should read: Acad Med. 2020;95(4):582-589.

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